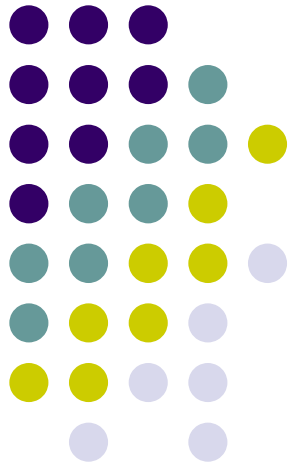
Chapter 25

Current And Resistance



PHYS 2321
Week 7: Current and Resistance

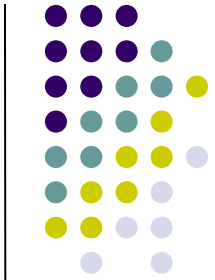
W7 Day 2 Outline

1) Quiz on capacitance
2) Ch. 25 Electric current and resistance
   Average and instantaneous current
   Drift velocity

Ch. 25 P. 2,3,6,9,12-14,16,19,28,29,33,47,55,56
Read 25 sec 1-5,7,8
Ch. 26 MisQ 1-9 odd
Read 26-1 to 26-4 Due next Wed.

Notes:
See test bank practice quiz on Ch. 26.
There will be more problems assigned from Ch. 26.

Electric Current

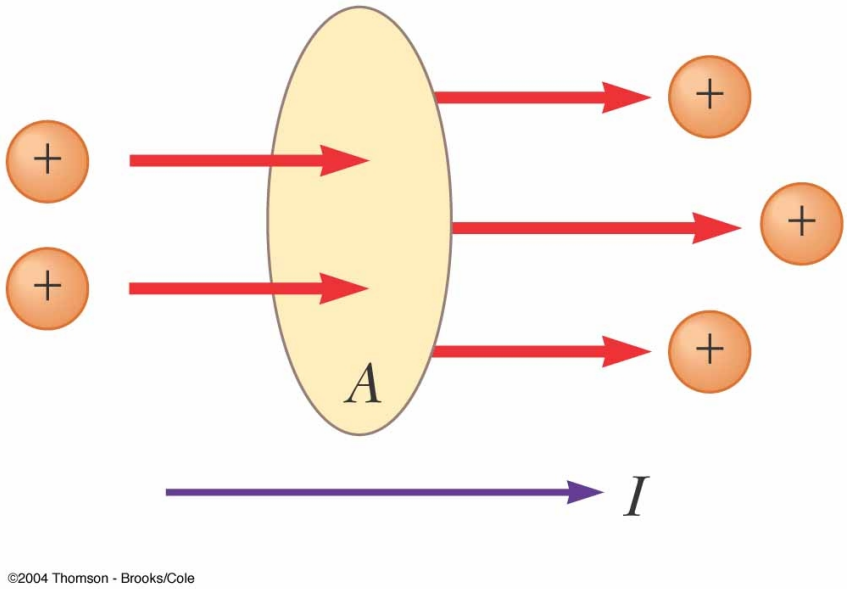


- **Electric current** is the rate of flow of charge through some region of space
- The SI unit of current is the **ampere** (A)
  - 1 A = 1 C / s
- The symbol for electric current is I

Average Electric Current

- Assume charges are moving perpendicular to a surface of area *A*
- If *Q* is the amount of charge that passes through *A* in time Δt, then the *average current* is

$$I_{avg} = \frac{Q}{\Delta t}$$



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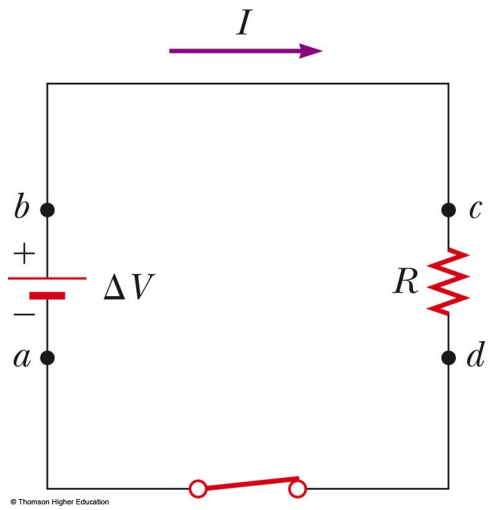
Current density, j = I/A

Instantaneous Electric Current

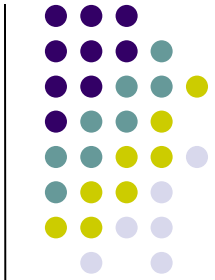
- If the rate at which the charge flows varies with time, the instantaneous current, I, can be found

$$I = \frac{dq}{dt}$$

- "Ohm's law" V=IR



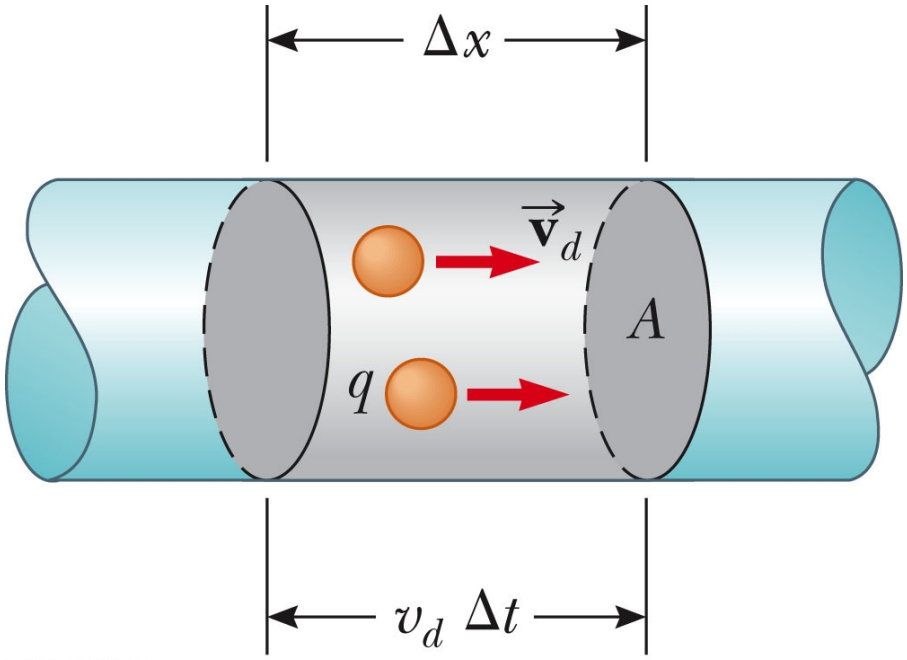
Direction of Current



- The charges passing through the area could be positive or negative or both
- It is conventional to assign to the current the same direction as the flow of positive charges
- The direction of current flow is opposite the direction of the flow of electrons
- It is common to refer to any moving charge as a *charge carrier*

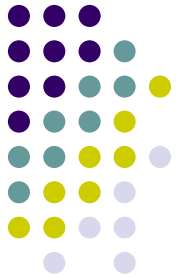
Current and Drift Speed

- Charged particles move through a conductor of cross-sectional area *A*
- *n* is the number of charge carriers per unit volume
- *nA*Δ*x* is the total number of charge carriers in gray region



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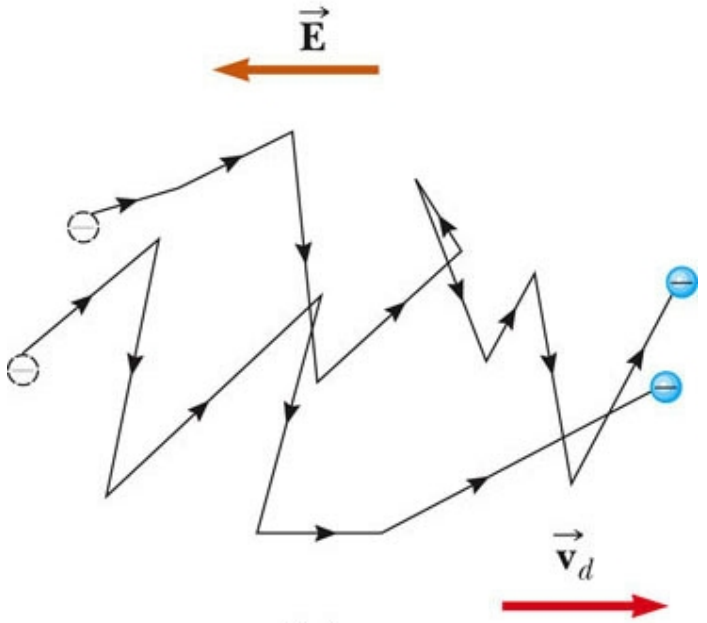
Current and Drift Speed, cont



- The total charge is the number of carriers times the charge per carrier, q
  - $\Delta Q = (nA\Delta x)q$
- The drift speed, v_d, is the *average* speed at which the carriers move
  - $v_d = \Delta x / \Delta t$ and $\Delta x = v_d \Delta t$
- Rewritten: $\Delta Q = (nAv_d \Delta t)q$
- Finally, current, $I_{ave} = \Delta Q/\Delta t = nqv_dA$

Charge Carrier Motion in a Conductor

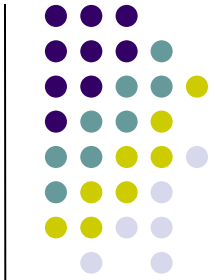
- The zigzag black lines represents the motion of a charge carrier in a conductor
  - The net drift speed is small
- The sharp changes in direction are due to collisions
- The net motion of electrons is opposite the direction of the electric field



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PLAY ACTIVE FIGURE

PHYS 2321
Week 7: Current and Resistance



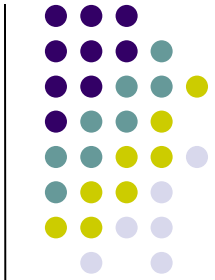
W7 Day 3 Outline

1) Return Quiz 3 (m=6/8,6.9/9) and Ch. 24 hwk.
2) Ch. 25 Electric current and resistance
   Drift velocity example. $Q = \int I\,dt$ example.
   Resistance and resistivity

Ch. 25 P. 2,3,6,9,12-14,16,19,28,29,33,47,55,56
Read 25 sec 1-5,7,8
Ch. 26 MisQ 1-9 odd
Read 26-1 to 26-4 Due next Wed.

Notes:
Ch. 24 hwk (means=9.68/10, 9.85/10)
There will be more problems assigned from Ch. 26.
Public Event at observatory 8-10 pm ?

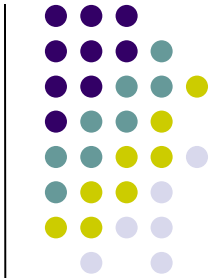
Drift Velocity, Example



- Assume a copper wire, with one free electron per atom contributed to the current
- The drift velocity for a 12-gauge copper wire carrying a current of 10.0 A is

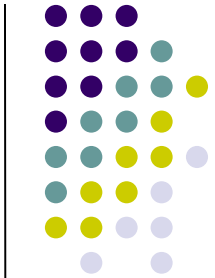
  2.23×10^{-4} m/s
  - This is the typical order of magnitude for drift velocities

Motion of Charge Carriers, final



- The electrons are already in the wire
- They respond to the electric field set up by the battery
- The battery does not supply the electrons, it only establishes the electric field

Some Final Notes About Current



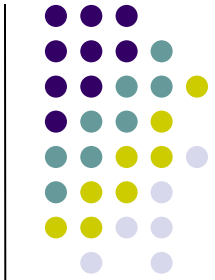
- A single electron is moving at the drift velocity in the circuit
  - It may take hours for an electron to move completely around a circuit
- The current is the same everywhere in the circuit
  - Current is not “used up” anywhere in the circuit
- The charges flow in the same rotational sense at all points in the circuit

Resistance

- In a conductor, the voltage applied across the ends of the conductor is proportional to the current through the conductor
- The constant of proportionality is called the **resistance** of the conductor

$$R = \frac{\Delta V}{I}$$

Resistance, cont.



- SI units of resistance are *ohms* (Ω)
  - 1 Ω = 1 V / A
- Resistance in a circuit arises due to collisions between the electrons carrying the current with the fixed atoms inside the conductor

PHYS 2321

Week 8: Current and Resistance, DC Circuits

W8 Day 1 (Wed) Outline

1) Ch. 25 Electric current and resistance
   $R = \rho\, l/A$ example
   Temperature dependence of R and ρ
   Power, $P=IV = I^2R = V^2/R$

Ch. 25 P. 2,3,6,9,12-14,16,19,28,29,33,47,55,56
Read 25 sec 1-5,7,8
Ch. 26 MisQ 1-9 odd
Read 26-1 to 26-4 Due Today (<3 pm)
Ch. 26 P. 1,2,5,7,12,17,33 (31 optional, for more practice)
Due Monday. 5pts

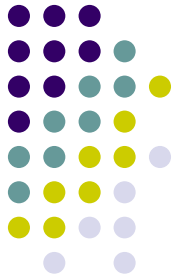
Notes: Pass back old homework

Resistivity

- The inverse of the conductivity is the **resistivity**:
  - $\rho = 1 / \sigma$
- Resistivity has SI units of ohm-meters ($\Omega \cdot m$)
- Resistance is a function of geometry and resistivity:

$$R = \rho \frac{l}{A}$$

Resistance and Temperature

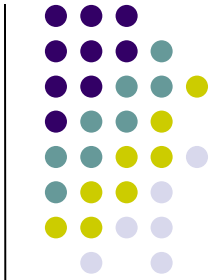


- Over a limited temperature range, the resistivity of a conductor varies approximately linearly with the temperature

$$\rho = \rho_O \left[1 + \alpha \left(T - T_0 \right) \right]$$

  - ρ_o is the resistivity at some reference temperature T_o
    - T_o is usually taken to be 20° C
    - α is the **temperature coefficient of resistivity**
      - SI units of α are $^oC^{-1}$

Temperature Variation of Resistance



- Since the resistance of a conductor with uniform cross sectional area is proportional to the resistivity, you can find the effect of temperature on resistance

$$R = R_o[1 + \alpha(T - T_o)]$$

- Use of this property enables precise temperature measurements through careful monitoring of the resistance of a probe made from a particular material

Resistivity Values

TABLE 27.2

Resistivities and Temperature Coefficients of Resistivity for Various Materials

Material	Resistivity[a] ($\Omega \cdot m$)	Temperature Coefficient[b] $\alpha[(^\circ C)^{-1}]$
Silver	1.59×10^{-8}	3.8×10^{-3}
Copper	1.7×10^{-8}	3.9×10^{-3}
Gold	2.44×10^{-8}	3.4×10^{-3}
Aluminum	2.82×10^{-8}	3.9×10^{-3}
Tungsten	5.6×10^{-8}	4.5×10^{-3}
Iron	10×10^{-8}	5.0×10^{-3}
Platinum	11×10^{-8}	3.92×10^{-3}
Lead	22×10^{-8}	3.9×10^{-3}
Nichrome[c]	1.50×10^{-6}	0.4×10^{-3}
Carbon	3.5×10^{-5}	-0.5×10^{-3}
Germanium	0.46	-48×10^{-3}
Silicon[d]	2.3×10^{3}	-75×10^{-3}
Glass	10^{10} to 10^{14}	
Hard rubber	$\sim 10^{13}$	
Sulfur	10^{15}	
Quartz (fused)	75×10^{16}	

[a] All values at 20°C. All elements in this table are assumed to be free of impurities.

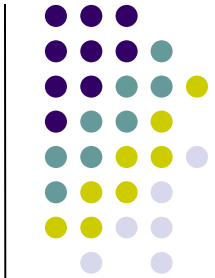
[b] See Section 27.4.

[c] A nickel–chromium alloy commonly used in heating elements.

[d] The resistivity of silicon is very sensitive to purity. The value can be changed by several orders of magnitude when it is doped with other atoms.

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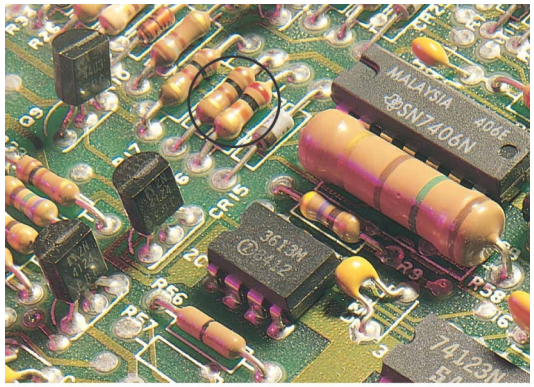
Resistance and temperature



Quick Quiz 27.4:
When does a light bulb with a metal filament carry more current,
a) immediately after it is turned on and the glow of the metal filament is increasing, or
b) after it has been on for a few milliseconds and the glow is steady?

Ans: a
Dimmer → cooler → lower R → higher I
Brighter→ hotter → higher R → lower I

Resistor Values (skip)



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- Values of resistors are commonly marked by colored bands

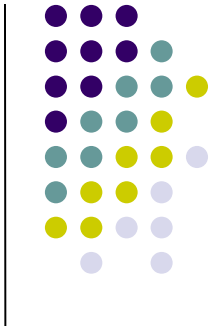
TABLE 27.1

Color Coding for Resistors

Color	Number	Multiplier	Tolerance
Black	0	1	
Brown	1	10^1	
Red	2	10^2	
Orange	3	10^3	
Yellow	4	10^4	
Green	5	10^5	
Blue	6	10^6	
Violet	7	10^7	
Gray	8	10^8	
White	9	10^9	
Gold		10^{-1}	5%
Silver		10^{-2}	10%
Colorless			20%

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Resistance and Resistivity, Summary



- Every ohmic material has a characteristic resistivity that depends on the properties of the material and on temperature
  - Resistivity is a property of substances
- The resistance of an object depends on its geometry and its resistivity
  - Resistance is a property of an object
- An ideal conductor would have zero resistivity
- An ideal insulator would have infinite resistivity

Current Density

- ***J*** is the **current density** of a conductor
- It is defined as the current per unit area
  - $\boldsymbol{J} = I / A = nq\boldsymbol{v}_d$
  - This expression is valid only if the current density is uniform and *A* is perpendicular to the direction of the current
- *J* has SI units of A/m^2
- The current density is in the direction of the positive charge carriers

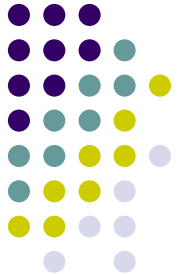
Conductivity

- A current density and an electric field are established in a conductor whenever a potential difference is maintained across the conductor
- For some materials, the current density is directly proportional to the field, $J = \sigma E$
- The constant of proportionality, σ, is called the **conductivity** of the conductor

Ohm's Law

- **Ohm's law** states that for many materials, the ratio of the current density to the electric field is a constant σ that is independent of the electric field producing the current
  - Most metals obey Ohm's law
  - Mathematically, $J = \sigma E$
  - Materials that obey Ohm's law are said to be *ohmic*
  - $I = V/R$ is often called "Ohm's Law" but objects will obey I=V/R even if they are not Ohmic! V/R must remained fixed over large range of I.

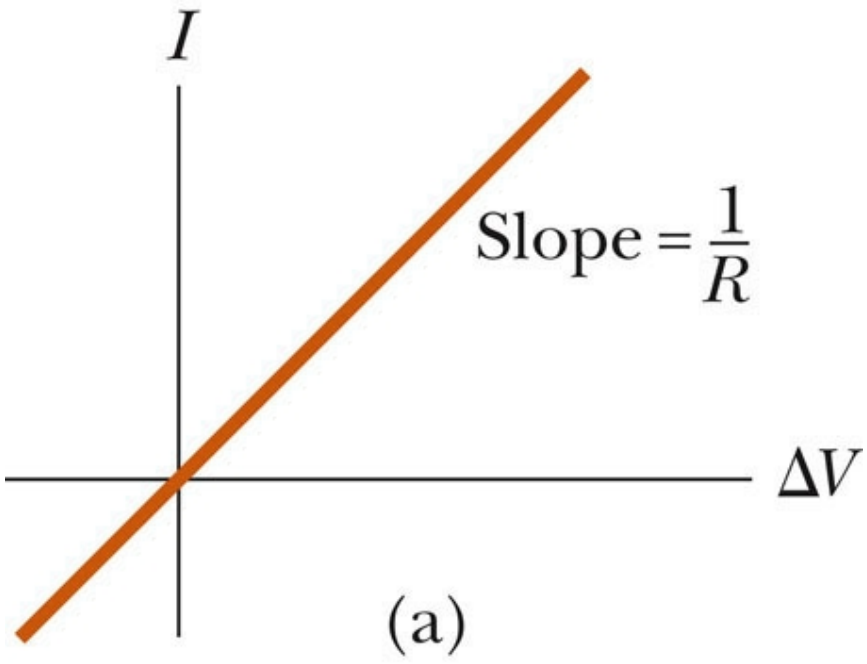
Ohm's Law, cont.



- Not all materials follow Ohm's law
  - Materials that do not obey Ohm's law are said to be *nonohmic*
- Ohm's law is not a fundamental law of nature
- Ohm's law is an empirical relationship valid only for certain materials over limited voltage ranges.

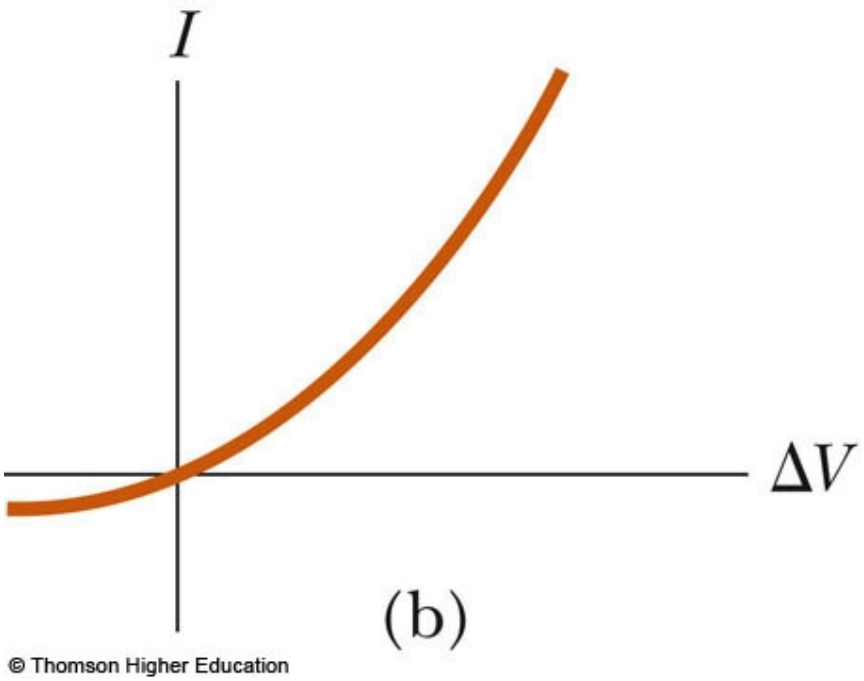
Ohmic Material, Graph

- An ohmic device
- The resistance is constant over a wide range of voltages
- The relationship between current and voltage is linear
- The slope is related to the resistance



Nonohmic Material, Graph

- Nonohmic materials are those whose resistance changes with voltage or current
- The current-voltage relationship is nonlinear
- A junction diode is a common example of a nonohmic device



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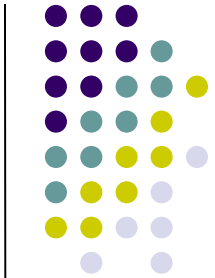
Georg Simon Ohm

- 1789 -1854
- German physicist
- Formulated idea of resistance
- Discovered the proportionalities now known as forms of Ohm's Law



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Electrical Conduction – A Model (skip)



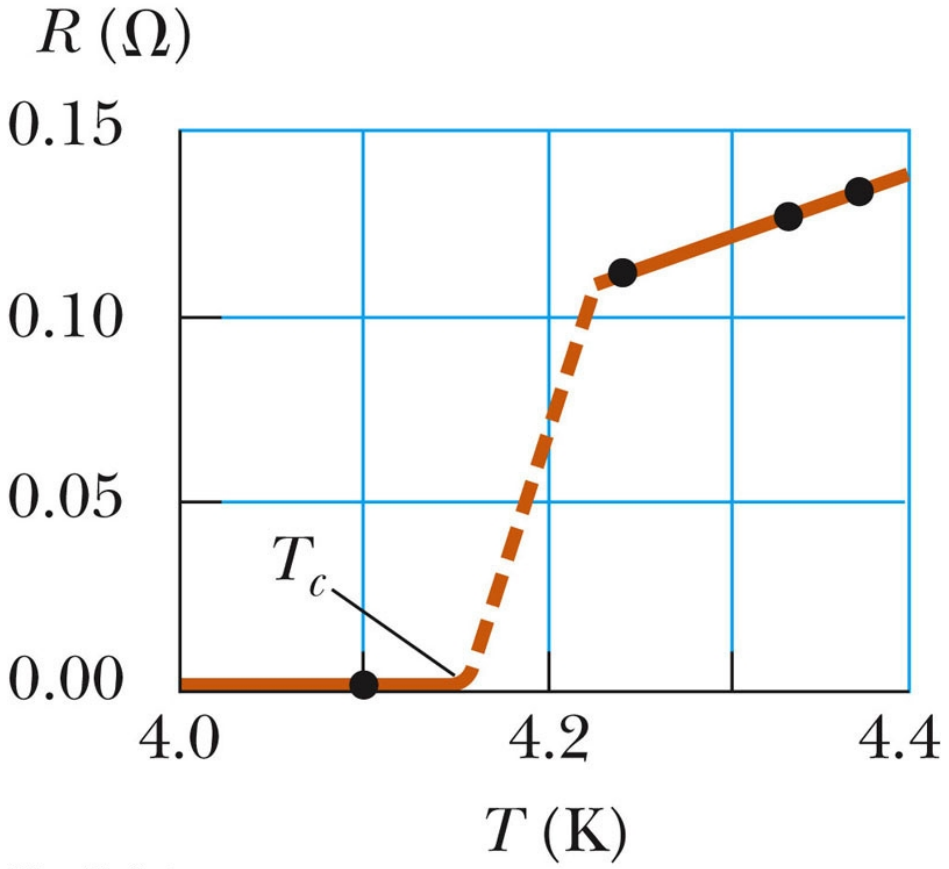
- Treat a conductor as a regular array of atoms plus a collection of free electrons
  - The free electrons are often called conduction electrons
  - These electrons become free when the atoms are bound in the solid
- In the absence of an electric field, the motion of the conduction electrons is random
  - Their speed is on the order of 10^6 m/s

Conduction Model, 2

- When an electric field is applied, the conduction electrons are given a drift velocity
- Assumptions
  - The electron's motion after a collision is independent of its motion before the collision
  - The excess energy acquired by the electrons in the electric field is lost to the atoms of the conductor when the electrons and atoms collide
    - This causes the temperature of the conductor to increase

Superconductors

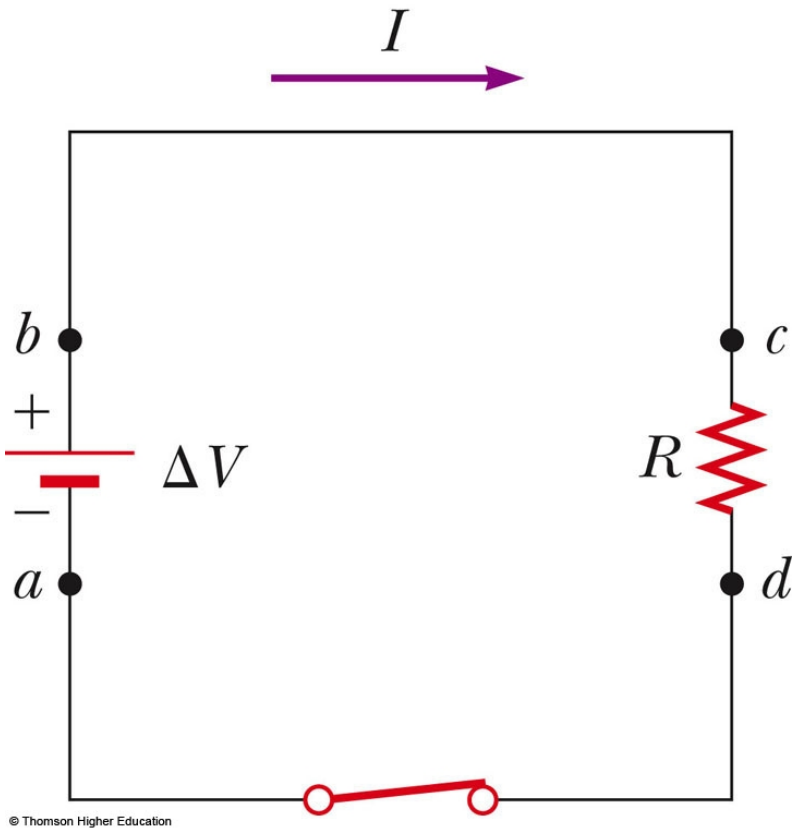
- A class of materials and compounds whose resistances fall to virtually zero below a certain temperature, T_C
  - T_C is called the **critical temperature**
- The graph is the same as a normal metal above T_C, but suddenly drops to zero at T_C



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Electrical Power

- Assume a circuit as shown
- As a charge moves from *a* to *b*, the electric potential energy of the charge increases by $q\Delta V$
  - The chemical energy in the battery must decrease by this same amount

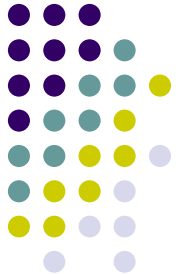


PLAY ACTIVE FIGURE

Electrical Power, 2

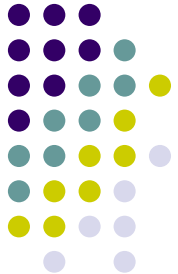
- As the charge moves through the resistor (*c* to *d*), the system loses this electric potential energy during collisions of the electrons with the atoms of the resistor
- This energy is transformed into internal energy in the resistor
  - Corresponds to increased vibrational motion of the atoms in the resistor

Electric Power, 3



- The resistor is normally in contact with the air, so its increased temperature will result in a transfer of energy by heat into the air
- The resistor also emits thermal radiation
- After some time interval, the resistor reaches a constant temperature
  - The input of energy from the battery is balanced by the output of energy by heat and radiation

Electric Power, 4



- The rate at which the system loses potential energy as the charge passes through the resistor is equal to the rate at which the system gains internal energy in the resistor
- The **power** is the rate at which the energy is delivered to the resistor

Electric Power, final

- The power is given by the equation:

$$P = I\Delta V$$

- Applying Ohm's Law, alternative expressions can be found:

$$P = I\Delta V = I^2 R = \frac{(\Delta V)^2}{R}$$

- Units: I is in A, R is in Ω, V is in V, and P is in W